

Isaac Sung

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SUMMARY

Software engineer with 6+ years of relevant professional experience building scalable web applications, mentoring cross-functional teams, and teaching computer science. Proven track record of improving user experience through data-driven experimentation, performance optimization, and collaborative product development. Passionate about creating customer-focused solutions that balance technical excellence with business impact.

WORK EXPERIENCE

NATIONAL CHRISTIAN FOUNDATION

Senior Software Engineer

Remote

May 2022 – Current

- Lead cross-functional development teams across 2 Agile squads, facilitating collaboration between API, data, UX design, and frontend teams to deliver customer-focused solutions.
- Architect and execute large-scale migration from legacy Salesforce systems to modern React tech stack, improving maintainability and developer productivity.
- Deliver significant performance improvements including 45% reduction in build times and 80% decrease in bundle size through strategic framework migration to TanStack Start.
- Collaborate closely with the UX design team to establish a standardized design system in Figma and implement a reusable component library used across product teams.
- Mentor team members on modern web development practices and led code reviews to maintain high code quality standards.
- Ship features using TypeScript, React, Node.js, TanStack Start, with continuous deployment via GitHub Actions and Azure Web Apps.

AUTOMATTIC, INC

Senior Software Engineer (Contractor)

Remote

Oct 2021 – Jan 2022

- Reduced user error rates by 21% through UX improvements to an A/B testing management web application, working closely with product stakeholders to identify pain points.
- Enhanced data discovery capabilities by migrating complex data grids from Material UI to AG Grid React, improving sorting and filtering performance for large datasets.
- Collaborated with product managers to prioritize feature development based on user feedback and usage analytics.

RIPON COLLEGE

Assistant Professor

Ripon, WI

Aug 2019 – Dec 2021

- Taught a variety of CS courses (using Python and C#) including Data Structures, Machine Learning, and Data Visualization.
- Led and mentored team of 3-4 undergraduate developers in designing and building an educational algebra game using Unity Engine, demonstrating project management and mentoring skills.
- Achieved 4.4/5.0 average rating from student evaluations through effective communication and hands-on learning approach.
- Developed a curriculum that balanced theoretical concepts with practical application, preparing students for industry roles.

MUSICNOTES, INC

Data Science Intern

Madison, WI

May 2018 – Aug 2018

- Built interactive web dashboards and data visualizations using JavaScript, Node, React, and D3, enabling stakeholders to make data-driven product decisions.
- Developed ML prototype to predict customer purchase behavior with ~80% accuracy using Python.

UNIVERSITY OF WISCONSIN-MADISON

Teaching Assistant

Madison, WI

Sep 2014 – May 2017

- Automated feedback systems by managing custom testing servers, providing students with instant assignment feedback and improving learning outcomes.
- Designed comprehensive programming assignments for an intro Python course and an intro C++ course, balancing educational value with practical application.
- Created robust testing frameworks by writing unit tests for Java course assignments, ensuring code quality and consistent grading.

EDUCATION

UNIVERSITY OF WISCONSIN–MADISON

Ph.D. in Computer Sciences and Education

Thesis: *Productive and Unproductive Frictions in Game Design*

Madison, WI

2016 – 2021

UNIVERSITY OF WISCONSIN–MADISON

Master's in Computer Sciences

Madison, WI

2014 – 2016

UNIVERSITY OF OKLAHOMA

Bachelor's in Computer Science

Norman, OK

2010 – 2014

PROJECTS

WEATHERGUESSER - TEMPERATURE GUESSING GAME

2021 – Current

- Hobby web game where players guess the current temperature of a random city from around the world.
- Implemented using TypeScript, React, TanStack Start, and OpenWeather API.

BRAVE NEW WORD - DISSERTATION WORD GAME

2016 – 2021

- Word search game developed with multiple UI/UX variations in order to research productive and unproductive frictions in game design.
- Implemented using Unity Engine, C#, and Firebase.

ADDITIONAL

Technical Skills: JavaScript, TypeScript, HTML/CSS, React, Node.js, Python, SQL, C#, Java, C++, PHP, Git, GitHub Actions, Jira/Atlassian, Azure, CI/CD

Certifications & Training: OWASP Top 10 Training